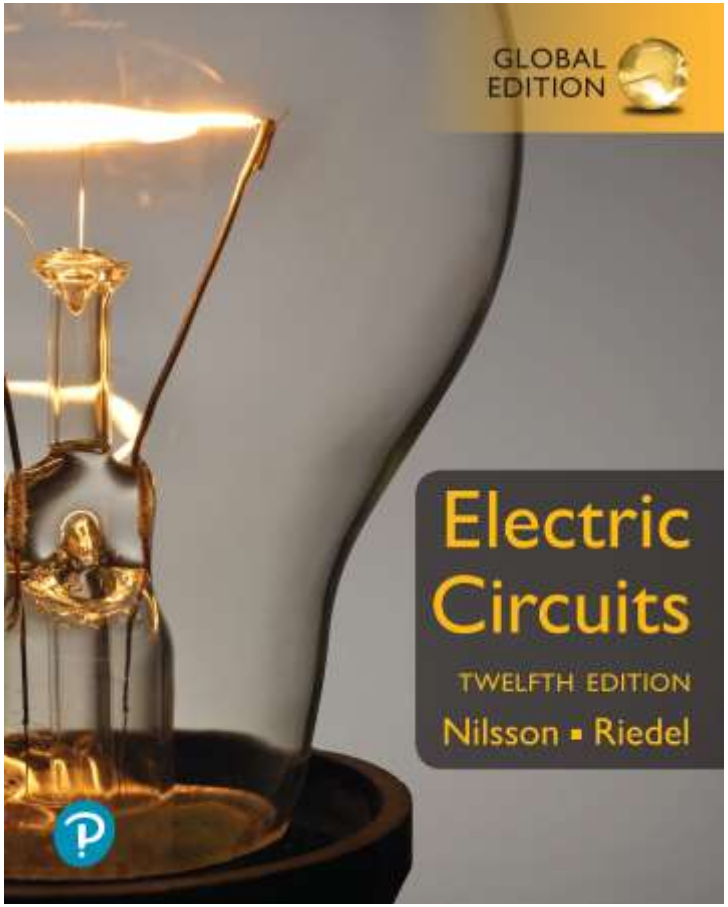
Electric Circuits

Twelfth Edition, Global Edition



Chapter 9 – Part 1

Sinusoidal Steady-State Analysis

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (1 of 24)

Objectives:

9.1 Be able to perform a phasor transform and its inverse;

9.2 Be able to phasor-transform a circuit;

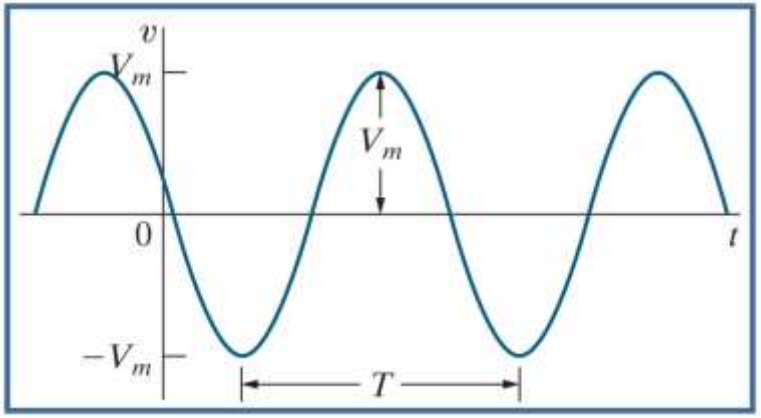
9.3 Solve arbitrarily complex circuits with sinusoidal sources using phasor method.

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (2 of 24)

Review the terminology of sinusoids



$$v(t) = V_m \cos(\omega t + \phi) \text{ V}$$

- V_m : Amplitude or magnitude
- ω : angular frequency [rad/s]
- t : time [s]
- ϕ : phase angle [deg]
- $f = \omega / 2\pi$: frequency [Hz]
- $T = 1/f$: period [s]

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (3 of 24)

What is the frequency of the sinusoid given below?

$i(t) = 36\cos(4\pi t + 45°)\,\text{mA}$

A. $4\pi t$

B. $4\pi\,Hz$

C. $4\pi\,rad/s$

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (3 of 24)

What is the frequency of the sinusoid given below?

$i(t) = 36\cos(4\pi t + 45°)\,\text{mA}$

✗ A. $4\pi t$

✗ B. $4\pi\,Hz$

✓ C. $4\pi\,rad/s$

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (4 of 24)

What is the phase angle of the sinusoid given below?

$v(t) = 50\cos(3000t - 60°)$ V

A. *−60 deg*

B. *−60 rad*

C. *60 deg*

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (4 of 24)

What is the phase angle of the sinusoid given below?

$v(t) = 50\cos(3000t - 60°)$ V

- ✔ A. *−60 deg*
- ✘ B. *−60 rad*
- ✘ C. *60 deg*

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (5 of 24)

Suppose we want to find the value of the sinusoidal current described below at $t = 5$ ms. When we substitute this value for t into the sinusoid, the argument for the cosine function has what units?

$i(t) = 0.2\cos(50\pi t + 45°)$ A

A. degrees

B. degrees

C. *A mix of degrees and radians*

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (5 of 24)

Suppose we want to find the value of the sinusoidal current described below at $t = 5$ ms. When we substitute this value for t into the sinusoid, the argument for the cosine function has what units?

$i(t) = 0.2\cos(50\pi t + 45°)$ A

- ✗ A. degrees
- ✗ B. degrees
- ✓ C. *A mix of degrees and radians*

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (6 of 24)

Evaluating a sinusoid at a specified time

$$i(0.005) = 0.2\cos(50\pi t + 45°)\ \text{A}$$

50π has the units rad/s

$50\pi t$ has the units radians

45 has the units degrees

Need to convert $50\pi t$ to degrees, or 45 to radians, and set your calculator appropriately! At **(t = 1s)**

$$2\pi/360° = 50\pi(0.005)/\text{x}° \rightarrow \text{x} = 45°$$

Thus, $i(0.005) = 0.2\cos(45° + 45°)\ \text{A}$

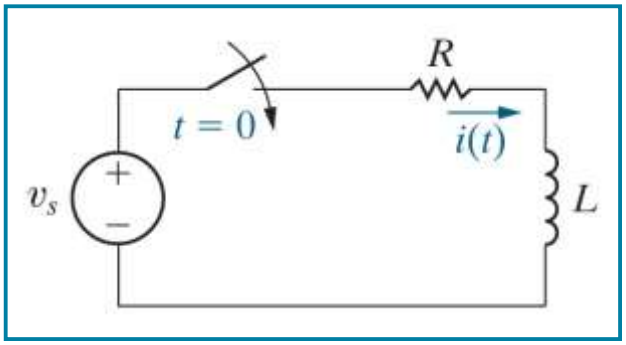
$$= 0.2\cos(90°) = 0\ \text{A}$$

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (7 of 24)

Suppose $v(t)=V_m\cos(\omega t+\phi)$ V in the circuit below:



The equation for *i(t)* is $L\dfrac{di(t)}{dt}+Ri(t)=V_m\cos(\omega t+\phi)$

The solution of this equation for *i(t)* is

$$i(t)=\frac{-V_m}{\sqrt{R^2+(\omega L)^2}}\cos(\phi-\theta)e^{-(R/L)t}+\frac{-V_m}{\sqrt{R^2+(\omega L)^2}}\cos(\omega t+\phi-\theta)$$

where $\theta=\tan^{-1}(\omega L/R)$

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (8 of 24)

Consider the equation for the current in the RL circuit with the sinusoidal source:

$$i(t) = \frac{-V_m}{\sqrt{R^2 + (\omega L)^2}} \cos(\phi - \theta) e^{-(R/L)t} + \frac{-V_m}{\sqrt{R^2 + (\omega L)^2}} \cos(\omega t + \phi - \theta)$$

Which term(s) go to 0 as $t \to \infty$?

A. The first term

B. The second term

C. Both terms

D. Neither term

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (8 of 24)

Consider the equation for the current in the RL circuit with the sinusoidal source:

$$i(t) = \frac{-V_m}{\sqrt{R^2 + (\omega L)^2}} \cos(\phi - \theta) e^{-(R/L)t} + \frac{-V_m}{\sqrt{R^2 + (\omega L)^2}} \cos(\omega t + \phi - \theta)$$

Which term(s) go to 0 as $t \to \infty$?

- ✔ A. The first term
- ✘ B. The second term
- ✘ C. Both terms
- ✘ D. Neither term

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (9 of 24)

$$i(t) = \frac{-V_m}{\sqrt{R^2 + (\omega L)^2}} \cos(\phi - \theta) e^{-(R/L)t} + \frac{-V_m}{\sqrt{R^2 + (\omega L)^2}} \cos(\omega t + \phi - \theta)$$

- The first term is the transient term – it decays to 0 as *t* goes to infinity
- The second term is the steady-state term – it persists for all time greater than 0. This term
  - Is sinusoidal
  - Has the same frequency as the input voltage
  - Has a different magnitude and phase angle compared to the input voltage

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (10 of 24)

- The circuit analysis techniques we are studying will allow us to calculate the sinusoidal steady-state response of the circuit – that is, the circuit's response to a sinusoidal input once the transient response has effectively decayed to 0.
- This is also called the AC Steady-State (ACSS) response.

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (11 of 24)

- Phasor
  - A complex number in polar form, with a magnitude and phase angle
  - Derived from a sinusoid using the phasor transform

Euler's identity: $e^{\pm j\theta} = \cos\theta \pm j\sin\theta$

$\Rightarrow \quad \cos\theta = \mathcal{Re}\left\{e^{j\theta}\right\}$

Now $v(t) = V_m \cos(\omega t + \phi)$

$$= V_m \mathcal{Re}\left\{e^{j(\omega t+\phi)}\right\}$$

$$= V_m \mathcal{Re}\left\{e^{j\omega t} e^{j\phi}\right\}$$

$$= \mathcal{Re}\left\{V_m e^{j\omega t} e^{j\phi}\right\}$$

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (12 of 24)

- Phasor transform
  - Extracts a sinusoid's magnitude and phase angle
  - Transforms a function of time into a function of frequency

$$\boldsymbol{V} = \mathcal{P}\left\{V_m \cos(\omega t + \phi)\right\} = V_m e^{j\phi} = V_m \angle \phi$$

- Inverse phasor transform
  - Turns a phasor back into a sinusoid, if someone tells you the frequency

$$v(t) = \mathcal{P}^{-1}\left\{\mathbf{V}\right\} = \mathcal{P}^{-1}\left\{V_m \angle \phi\right\} = V_m \cos(\omega t + \phi)$$

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (13 of 24)

Suppose

$$i(t) = 36\cos(4\pi t + 45°)\ mA$$

Then the phasor transform of $i(t)$ is

A. 36 mA

B. $36\angle 45°$ mA

C. $36\angle 4\pi t$ mA

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (13 of 24)

Suppose

$$i(t) = 36\cos(4\pi t + 45°)\ mA$$

Then the phasor transform of $i(t)$ is

- ✗ A. 36 mA
- ✓ B. $36\angle 45°$ mA
- ✗ C. $36\angle 4\pi t$ mA

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (14 of 24)

Adding or subtracting sinusoids in the time domain is hard (you need trig identities!) But if you phasor-transform the sinusoids, it is easy to combine them. For example,

$$i(t) = \left[5\cos(300t + 36.87°) + 10\cos(300t - 53.13°)\right] \; A$$

$$\boldsymbol{I} = 5\angle 36.87° + 10\angle -53.13° = 11.18\angle -26.57° \; A$$

$$\therefore \qquad i(t) = \mathcal{P}^{-1}\left\{\boldsymbol{I}\right\} = 11.18\cos(300t - 26.57°) \; A$$

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (15 of 24)

Consider the equations that relate voltage and current in a resistor, inductor, and capacitor in the time domain, and in the phasor domain.

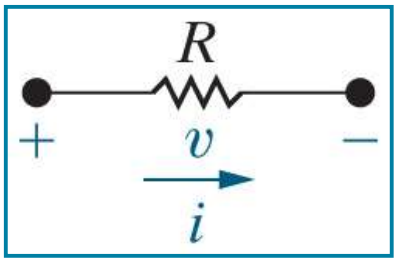
Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (16 of 24)

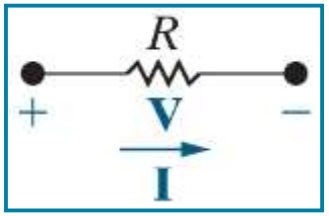
Resistors:

Time domain



$\mathcal{P}\rightarrow$

Phasor (frequency) domain



$$i(t) = I_m \cos(\omega t + \theta)$$

$$\therefore\ v(t) = Ri(t) = RI_m \cos(\omega t + \theta)$$

$$\boldsymbol{I} = I_m \angle \theta$$

$$\therefore \quad \boldsymbol{V} = R\boldsymbol{I} = RI_m \angle \theta$$

Note that the voltage and current phasors for a resistor have the same phase angle – therefore we say that the voltage and current in a resistor are “in phase”.

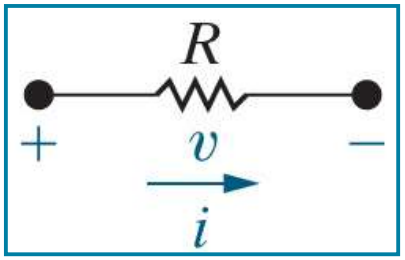
Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (17 of 24)

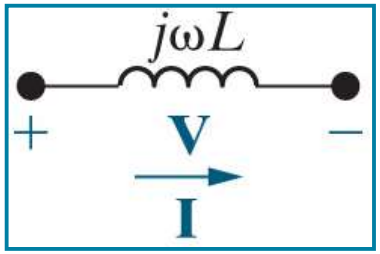
Inductors:

Time domain



$\mathscr{P}\rightarrow$

Phasor (frequency) domain



$$i(t) = I_m \cos(\omega t + \theta)$$

$$\therefore \quad v(t) = L\, di(t)/dt$$

$$= -\omega L I_m \sin(\omega t + \theta)$$

$$= -\omega L I_m \cos(\omega t + \theta - 90°)$$

$$\boldsymbol{I} = I_m \angle \theta$$

$$\therefore \quad \boldsymbol{V} = -\omega L I_m e^{j(\theta - 90°)}$$

$$= -\omega L I_m e^{j\theta} e^{-j90°}$$

$$= -\omega L I_m e^{j\theta}(-j)$$

$$= j\omega L I_m e^{j\theta} = j\omega L \boldsymbol{I}$$

In an inductor, the voltage leads the current by 90°

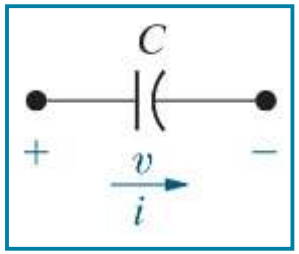
Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (18 of 24)

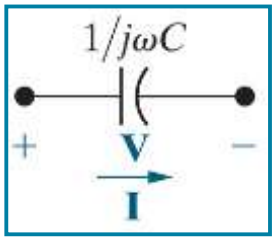
Capacitors:

Time domain



$\mathscr{P}\rightarrow$

Phasor (frequency) domain



$$v(t) = V_m \cos(\omega t + \theta)$$

$$\therefore \quad i(t) = C\,dv(t)/dt$$

$$= -\omega C V_m \sin(\omega t + \theta)$$

$$= -\omega C V_m \cos(\omega t + \theta - 90°)$$

$$\boldsymbol{V} = V_m \angle \theta$$

$$\therefore \quad \boldsymbol{I} = -\omega C V_m e^{j(\theta - 90°)}$$

$$= -\omega C V_m e^{j\theta} e^{-j90°}$$

$$= -\omega C V_m e^{j\theta}(-j)$$

$$= j\omega C V_m e^{j\theta} = j\omega C \boldsymbol{V}$$

In a capacitor, the voltage lags the current by 90°

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (19 of 24)

The impedance of a resistor is

A. Always a real number

B. Always an imaginary number

C. A complex number with real and imaginary parts

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (19 of 24)

The impedance of a resistor is

- ✓ A. Always a real number
- ✗ B. Always an imaginary number
- ✗ C. A complex number with real and imaginary parts

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (20 of 24)

The impedance of an inductor is

A. A real number

B. A positive imaginary number

C. A negative imaginary number

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (20 of 24)

The impedance of an inductor is

- ✗ A. A real number
- ✓ B. A positive imaginary number
- ✗ C. A negative imaginary number

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (21 of 24)

The impedance of a capacitor has the units

A. Farads [F]

B. Henries [H]

C. Ohms $[\Omega]$

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (21 of 24)

The impedance of a capacitor has the units

- ✗ A. Farads [F]
- ✗ B. Henries [H]
- ✓ C. Ohms [Ω]

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (22 of 24)

Impedance is a phasor.

A. True

B. False

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (22 of 24)

Impedance is a phasor.

✗ A. True

✓ B. False

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (23 of 24)

Summary:

- In the time domain
  - Resistor: $v(t) = Ri(t)$
  - inductor: $v(t) = L di(t)/dt$
  - Capacitor: $i(t) = C dv(t)/dt$
- In the phasor domain $\mathbf{V} = Z\mathbf{I}$
  - Z is impedance, defined as the ratio of V to I
  - Z has the units Ohms $[\Omega]$
  - Resistor: $Z_R = R$
  - Inductor: $Z_L = j\omega L$
  - Capacitor: $Z_C = 1/j\omega C = -j/\omega C$

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Chapter 9 – Sinusoidal Steady-State Analysis (24 of 24)

Summary:

- In the time domain
  - Ohm's law (only for resistors): $v = Ri$
  - KVL (around a loop): $v_1 + v_2 + \ldots + v_n = 0$
  - KCL (at a node): $i_1 + i_2 + \ldots + i_n = 0$
  - These three laws lead to all other time-domain circuit analysis techniques
- In the phasor domain
  - Ohm's law (for R, L, C): $V = ZI$
  - KVL (around a loop): $V_1 + V_2 + \ldots + V_n = 0$
  - KCL (at a node): $I_1 + I_2 + \ldots + I_n = 0$
  - These three laws mean we can use all time-domain circuit analysis techniques in the phasor domain!

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved

Copyright



This work is protected by United Kingdom copyright laws and is provided solely for the use of instructors in teaching their courses and assessing student learning. Dissemination or sale of any part of this work (including on the World Wide Web) will destroy the integrity of the work and is not permitted. The work and materials from it should never be made available to students except by instructors using the accompanying text in their classes. All recipients of this work are expected to abide by these restrictions and to honor the intended pedagogical purposes and the needs of other instructors who rely on these materials.

© 2024 Pearson Education Ltd. All Rights Reserved